

IMO PHASE II LEVEL 2

BIJECTIVE PROOF

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Bijective Proof: The main idea is similar to Double Counting: we are interested in finding a different (easier) way to count the required quantity.

Example 1. Let n be a positive integer. Determine the number of lattice paths from $(0, 0)$ to (n, n) using only unit up and right steps, such that the path stays in the region $x \geq y$.

Ans. It is easy to see that the total number of lattice paths from $(0, 0)$ to (n, n) without the $x \geq y$ restriction is $\binom{2n}{n}$. Let us count the number of paths that goes into the $x < y$ region. Call these paths bad paths. We are going to prove there is a bijection between bad paths and paths from $(-1, 1)$ to (n, n) , hence the total number of good paths equal

$$\binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1} \binom{2n}{n}.$$

Suppose that P is a bad path. Since P goes into the region $x < y$, it must hit the line $y = x + 1$ at some point. Let X be the first point on the path P that lies on the line $y = x + 1$. Now, reflect the portion of path P up to X about the line $y = x + 1$, keeping the latter portion of P the same. This gives us a new path P' starting from $(-1, 1)$.

For the inverse construction, for any path from $(-1, 1)$ to (n, n) , let X be the first point on the path lies on the line $y = x + 1$, and let Q' be constructed from Q by reflecting the first portion of Q up to X through the line $y = x + 1$ and keeping the rest the same. Then the inverse of the bijection given above sends Q to Q' .

The remaining part is to prove these two constructions are of an inverse relation. We will leave this as a simple exercise.

Example 2. For $n \geq 1$, prove that

$$\sum_{k=0}^n (-1)^k \binom{n}{k} = 0.$$

Ans. Take the elements of $[n]$. Take the element 1 (we know that $n \geq 1$). For every subset S of $[n]$ with k elements, where k is even, we can get a unique subset with an odd number of elements as follows: if $1 \in S$ remove it, otherwise, add it. And obviously, every subset with an odd number of element arise from this way. Hence there is a bijection between even subsets and odd subsets.

Problem 1. For $n \geq 0$, prove that

$$\sum_{k=0}^n \binom{x}{k} \binom{y}{n-k} = \binom{x+y}{n}.$$

Problem 2. A composition of n is a sequence $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k)$ of positive integers such that $\sum \alpha_i = n$. Proof that

1. the number of compositions of n is 2^{n-1} ,
2. the total number of parts of all compositions of n is equal to $(n+1)2^{n-2}$, and
3. for $n \geq 2$, the number of compositions of n with an even number of even parts is equal to 2^{n-2} .

Problem 3. The Fibonacci numbers F_n are defined by $F_1 = F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 2$. Prove that

1. the number $f(n)$ of compositions of n with parts 1 and 2 is F_{n+1} ,
2. the number of compositions of n with all parts > 1 is F_{n-1} , and
3. the number of compositions of n with odd parts is F_n .

Problem 4. Fix positive integers n and k . Find the number of k -tuples $(S_1, S_2, \dots, S_k)$ of subsets of S_i of $\{1, \dots, n\}$ for each of the following conditions separately:

1. $S_1 \subseteq S_2 \subseteq \dots \subseteq S_k$,
2. The S_i 's are pairwise disjoint,
3. $S_1 \cap S_2 \cap \dots \cap S_k = \emptyset$, and
4. $S_1 \subseteq S_2 \supseteq S_3 \subseteq S_4 \supseteq S_5 \subseteq \dots$.

Problem 5. Find the number of subsets of S of $[n]$ that do not contain two consecutive integers.

Problem 6. Find the number of binary sequences $(\varepsilon_1, \dots, \varepsilon_n)$ satisfying

$$\varepsilon_1 \leq \varepsilon_2 \geq \varepsilon_3 \leq \varepsilon_4 \geq \varepsilon_5 \leq \dots.$$

Problem 7. For $n \geq 0$, prove that

$$\sum_{k=0}^n \binom{x+k}{k} = \binom{x+n+1}{n}.$$

Problem 8. Let p be a prime. Then $\binom{2p}{p} - 2$ is divisible by p^2 .

Problem 9. Show that

$$\sum (2^{a_1-1} - 1) \dots (2^{a_k-1} - 1) = F_{2n-2},$$

where the sum is over all compositions $a_1 + a_2 + \dots + a_k = n$.